

METHOD



A novel method for EKG anomaly detection based on the double sliding window technique

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ABSTRACT

Traditional detection methods often rely on fixed thresholding or machine-learning models, which can be computationally expensive. This study introduces a double-sliding-window technique to detect anomalies in ECG signals through adaptive-signal analysis. This method uses two independent sliding windows to dynamically track signal variations, enabling real-time identification of deviations in heart rhythm without prior knowledge of data distribution. The detection module achieved an average execution-time of 0.03 seconds with using less than 16 MB of memory, supporting its suitability for real-time systems. Furthermore, confusion matrix analysis yielded an accuracy of 95.33%, a sensitivity of 95.00%, a precision of 100.00%, and a specificity of 100.00%.

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1. Introduction

Fault detection is an effective strategy to identify deviations in periodic signals that often compromise system integrity (Chandola *et al.* 2009). This task acquires particular relevance within biomedical applications, where numerous physiological signals exhibit periodic characteristics encoding essential diagnostic information (Bera *et al.* 2005; Patel *et al.* 2021). A representative case is the electrocardiogram (ECG), extensively utilized to assess cardiac activity (Kachuee *et al.* 2018). Aberrations in EKG signals, such as arrhythmic patterns, episodes of bradycardia, or morphological distortions, frequently serve as early markers of potentially life-threatening cardiovascular conditions (Al-Shrouf 2020; Kulanuwat *et al.* 2021; Patel *et al.* 2021). Detecting these conditions requires clinical evaluation in low-risk scenarios or immediate medical intervention in more severe cases (Kachuee *et al.* 2018).

Previous work relies on very complex methods, as artificial intelligence (AI) to supervise if the EKG signal falls beyond a threshold value (Kachuee *et al.* 2018; Mule *et al.* 2025; Bagul *et al.* 2026; Malave and Kaneshiro 2026). Even when the methodology that supervises the behavior of the biological signal can be

cutting edge, they depend on a fixed threshold that does not take into account variations in the baseline for the comparison. A fixed threshold makes the system very sensitive to noise: it can be triggered because of a healthy increase in cardiac rhythm due to an increase in physical activity of the patient, as well as for a sudden cardiac event as tachycardia.

To solve the previous problem, the proposed methodology presents a novel fault detection approach based on the double sliding window (DSW) technique (Braverman and Ostrovsky 2010; Cedeño Pozo and Trujillo Codorníu 2015; Zhang *et al.* 2016). Window techniques are useful to detect anomalies in any periodic signal, however, as EKG signals are quasi-periodic, segmentation for local linearization must be used. By comparing the behavior of the biological signal within a local interval, the baseline for the comparison will take noise as part of the new reality at the exact time of the comparison. The result is a technique that performs a series of dynamic comparisons between consecutive signal segments to evaluate the signal behavior in real-time, without relying on fixed thresholds. This dynamic adaptiveness will shift the comparison baseline according to the intrinsic variability of biomedical signals (Bera *et al.* 2005; Patel *et al.* 2021) to detect variations in frequency,

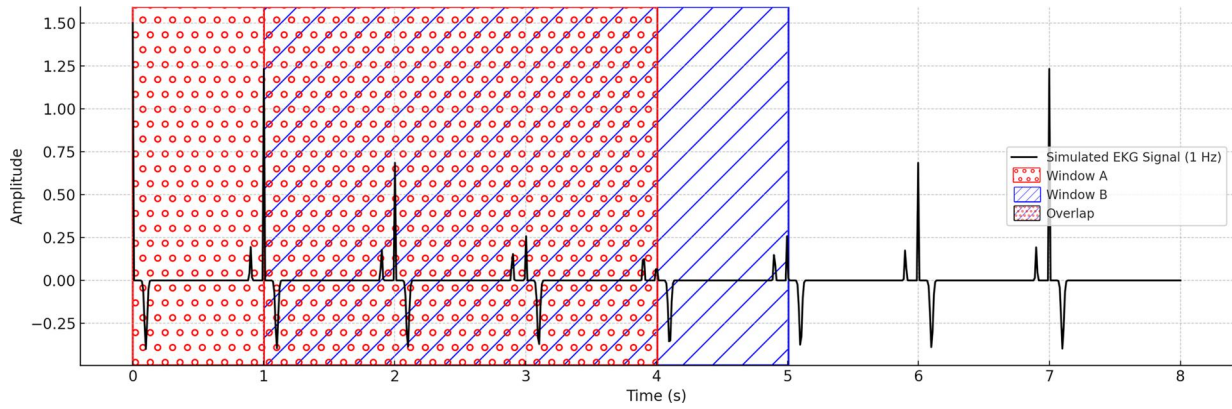


Figure 1. Illustration of the double sliding window approach applied to a simulated 1 Hz EKG signal.

amplitude, waveform shape (as measured by the area under the curve), and correlation (Cedeño Pozo and Trujillo Codorníu 2015; Desale and Shinde 2023). This way, the approach presented in this paper offers a flexible methodology to assess parameters such as sensitivity and specificity (Braverman and Ostrovsky 2010; Al-Shrouf 2020; Kulanuwat *et al.* 2021; Desale and Shinde 2023).

Data for the experiments in this paper were obtained from publicly available sources, including the PhysioNet MIT-BIH Arrhythmia Database (Goldberger *et al.* 2000) and preprocessed versions obtained *via* Kaggle (Kaggle 2025). Some tests used original, unmodified recordings to evaluate performance under realistic clinical conditions. Other experiments used these real signals as a base to generate controlled anomalies by modifying properties such as frequency, amplitude, and waveform shape. This strategy allowed systematic evaluation of the detection algorithm's sensitivity to diverse types and intensities of signal deviations.

Beyond their effectiveness and adaptability, DSW is computationally efficient, reducing the cost of biomedical signal analysis (Kehtarnavaz 2004; Bayvel 2013; Cedeño Pozo and Trujillo Codorníu 2015; Aishah 2024). This efficiency stems from their ability to process the signal in small overlapping segments (segmentation) rather than relying on access to the entire waveform (Braverman and Ostrovsky 2010; Cedeño Pozo and Trujillo Codorníu 2015). As a result, the detection module achieves fast execution times with minimal memory usage, meeting the requirements of real-time systems where low latency and lightweight operation are essential (Kehtarnavaz 2004; Zhang *et al.* 2023; Aishah 2024). Through a comprehensive evaluation, this paper aims to demonstrate this approach's effectiveness and practical applicability in EKG fault detection, providing

valuable insights into its potential extension to broader signal-processing contexts.

2. Proposed method

The proposed method is based on the concept of a double sliding window, initially developed for data stream analysis (Braverman and Ostrovsky 2010; Cedeño Pozo and Trujillo Codorníu 2015; Zhang *et al.* 2016), where two windows examine consecutive segments as shown in Figure 1. In this work, the strategy was deliberately modified by introducing an overlap between the two windows to adapt it to the domain of periodic signal processing. This overlap allows a more continuous and sensitive comparison between signal segments, which is especially useful for detecting subtle irregularities in waveform behavior across cycles.

Each iteration of the method involves extracting key features from the signal within both windows and comparing them using correlation metrics and measures, such as the area under the curve. If the degree of dissimilarity between the two segments surpasses a predefined allowable error, the window under evaluation is flagged as potentially anomalous. This process enables the system to distinguish between minor fluctuations due to physiological variability and genuine deviations that could indicate a fault or irregularity in the signal.

The systematic displacement of the windows across the signal allows a consistent and efficient scanning process that does not require labeled data or predefined anomaly templates (Braverman and Ostrovsky 2010; Tamalu *et al.* 2023). By not relying on a similar threshold to tolerate baseline variations or typical noise, there is no need for filtering or processes that may introduce latency or distort relevant waveform details. As a result, the proposed method has a low

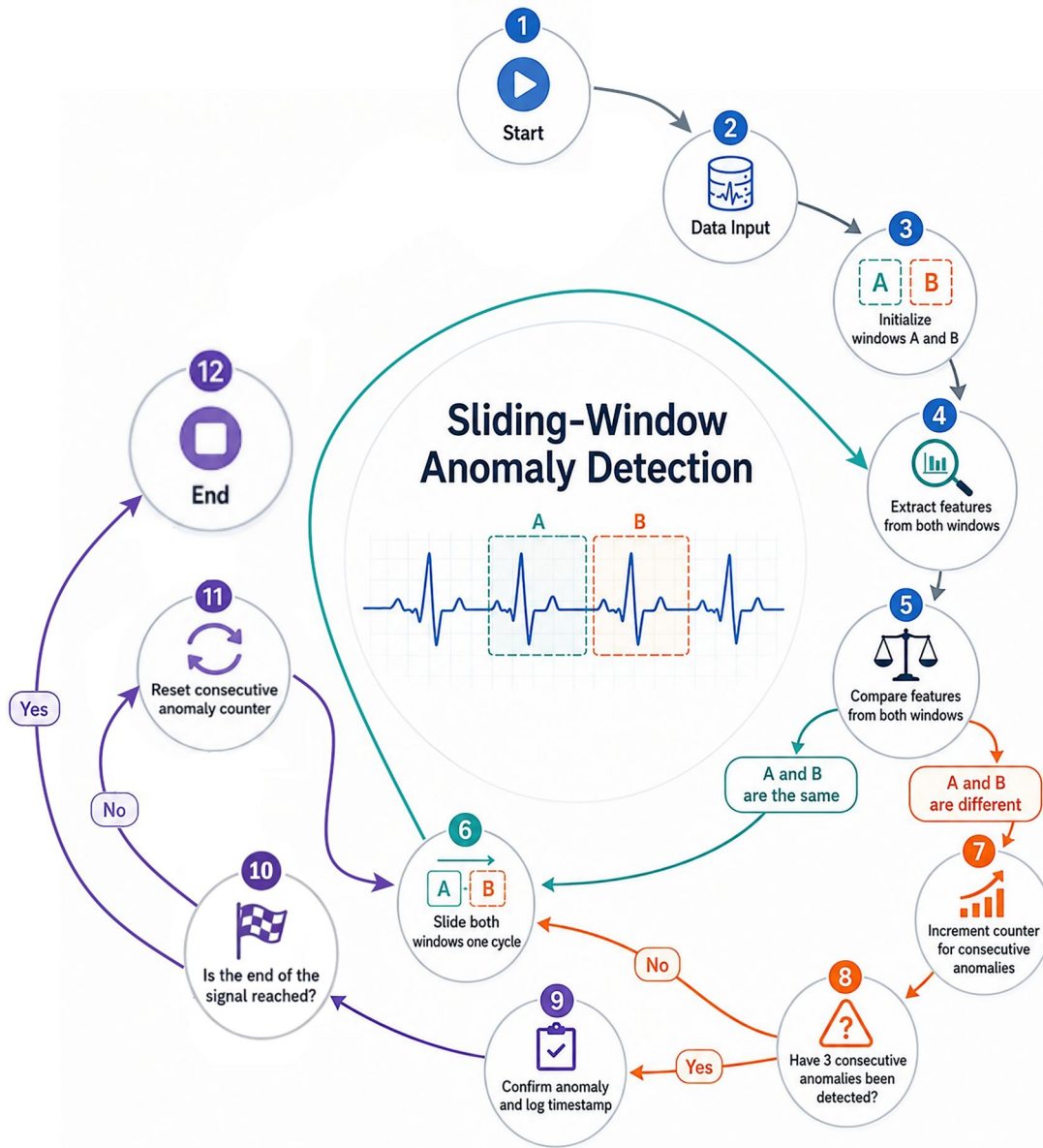


Figure 2. Flowchart of the DSW method for anomaly detection.

computational cost, making it suitable for a wide range of applications, including real-time (Kehtarnavaz 2004; Zhang *et al.* 2023; Aishah 2024).

Several experiments were conducted using electrocardiographic signals to evaluate the proposed method's effectiveness. These tests included cases where faults were manually introduced into otherwise normal EKG recordings, providing a comprehensive view of the method's ability to reliably identify abnormalities across different datasets.

3. The algorithm

The algorithm implemented in this study follows a structure specifically designed for detecting anomalies in periodic signals. This method operates in real-time,

continuously analyzing incoming data without interruptions or adaptive recalibration during execution. The fundamental idea behind this proposal is that deviations between overlapping segments of a periodic signal may indicate irregularities that compromise the underlying system behavior or signal integrity.

As shown in the flowchart in Figure 2, the algorithm starts with a procedure that estimates the frequency of the input signal. This estimated frequency provides the temporal reference needed to calculate the typical cycle duration used to determine the appropriate width of the sliding windows. According to this information, two windows are configured to cover four cycles (heartbeats) each. This window size will be sufficiently short to linearize the signal, but large enough to identify a deviation in one single

beat. The first window (A) is positioned one cycle ahead of the second window (B), resulting in an overlap, as shown in [Figure 1](#). This configuration ensures that each window provides sufficient temporal context to evaluate the waveform's evolution while maintaining high responsiveness to local changes.

The algorithm slides the windows along the data as the signal progresses, continuously comparing the information within them. Specifically, it evaluates a set of extracted characteristics from each window, including waveform correlation, energy distribution, amplitude variation, and potential frequency shifts. Comparing these metrics is essential to determine whether a significant deviation occurred between the two analyzed segments.

When a deviation exceeds a predefined set point, the algorithm does not immediately classify the segment as an anomaly; instead, it marks the current window as potentially anomalous. Suppose a predefined number of consecutive windows are marked as deviated. In that case, the algorithm confirms the presence of an anomaly and records the corresponding timestamp, associating it with the last window in the sequence. This persistence criterion strengthens the robustness of the detection strategy, ensuring that only consistent and sustained irregularities are flagged for further consideration. Once the anomaly is confirmed, the system may trigger secondary processes, such as alert generation, data labeling, or additional analysis modules.

Suppose a sliding window does not exhibit a significant deviation compared to the previous one. In that case, the anomaly counter is reset, and the algorithm continues to slide along the signal without interruption. This behavior allows the method to dynamically adapt to the natural variability of the signal without compromising precision or continuity. The process repeats until the signal's end is reached or a separate termination condition, such as a sustained zero input, is detected.

Overall, the algorithm balances sensitivity and specificity by combining a fixed-length window analysis with a decision based on consecutive confirmations. A reliable framework for real-time anomaly detection will be helpful for analysis in biomedical and other periodic signal environments.

4. Results and discussion

A standard EKG dataset was downloaded from the Kaggle platform to validate the proposed method. To emulate abnormal behaviors, this dataset was

manually tampered to generate approximately 300 EKG datasets featuring normal, slightly abnormal, and abnormal behaviors. The dataset encompasses diverse scenarios, from clean waveforms to sequences with systematically introduced anomalies. Controlled manipulations introduced these anomalies using an automated looping routine with magnitude and temporal position variations.

Once the datasets were prepared, the method was applied directly to the signals without preprocessing or filtering. As explained in the previous section, the segment size for the linearization of the signal was empirically settled as the time that takes 4 heartbeats to occur. This segmentation will be enough to adjust the baseline to include any noise in the system while maintaining accuracy. Both sliding windows will have a 75% overlap (3 periods, as illustrated in [Figure 1](#)). A deviation was confirmed as an anomaly when the correlation between the windows differed by more than 5% across three consecutive comparisons. This threshold was selected to balance responsiveness and robustness based on empirical analysis of preliminary tests with various settings.

In [Figure 3](#), an EKG signal with various intentionally introduced anomalies is shown to evaluate the performance of the proposed method. These include three consecutive abnormal cycles, a single irregular beat, and a segment with increased heart rate. Red dashed lines represent the timestamps identified by the algorithm, accurately marking the onset of each anomaly. The method successfully detects deviations in morphology and frequency, essential for capturing both transient and sustained irregularities. In contrast to purely statistical approaches, this method provides clear visual marks that strengthen interpretability and support precise human-in-the-loop diagnostics. Each detection is time-stamped, displayed as visual indicators on the signal plot, and printed in the command window, allowing easy documentation and further analysis.

A notable limitation of the method is the temporal resolution during real-time operation. Because anomaly confirmation requires three consecutive comparisons between windows in our tests, three cardiac cycles are needed after the anomaly has occurred to trigger an alert flag. This delay is further compounded by a modest computational latency, resulting in a slight, yet perceptible, lag in real-time applications.

Moreover, due to the structure of the sliding windows and the alignment strategy used, confirmed anomaly timestamps are registered at the beginning of the cycle where the anomaly is detected,

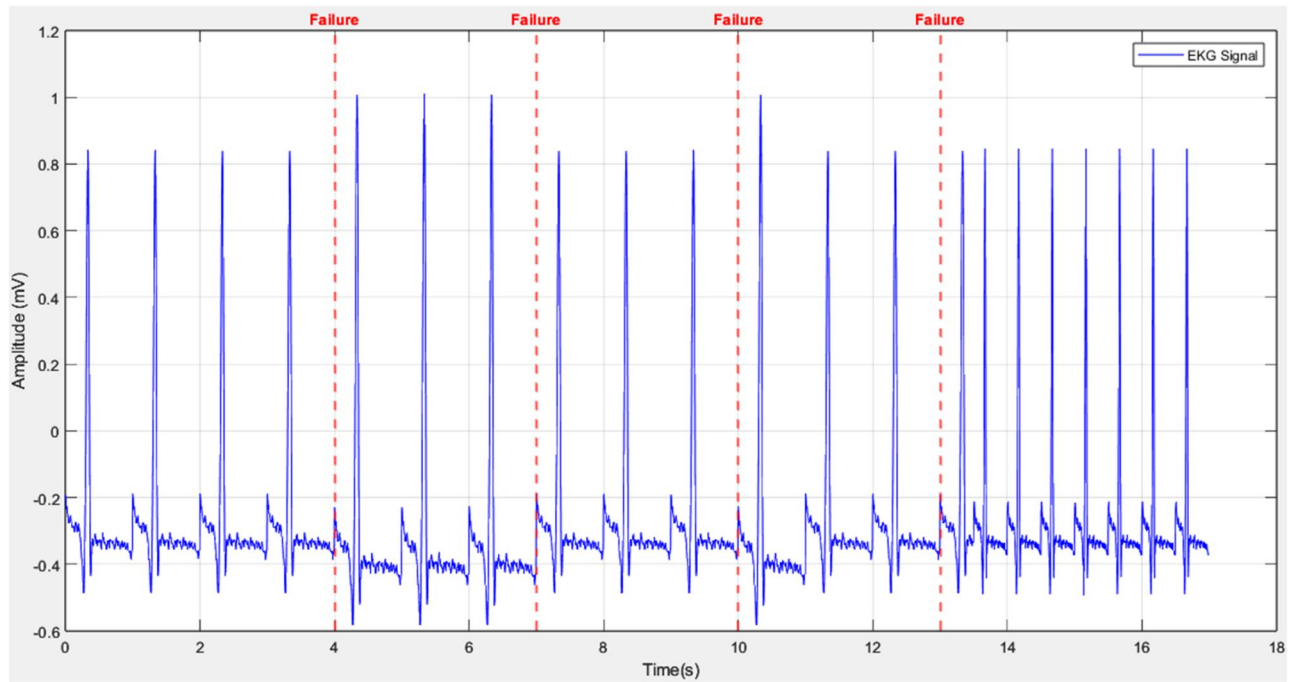


Figure 3. Fail detection results on a simulated EKG signal.

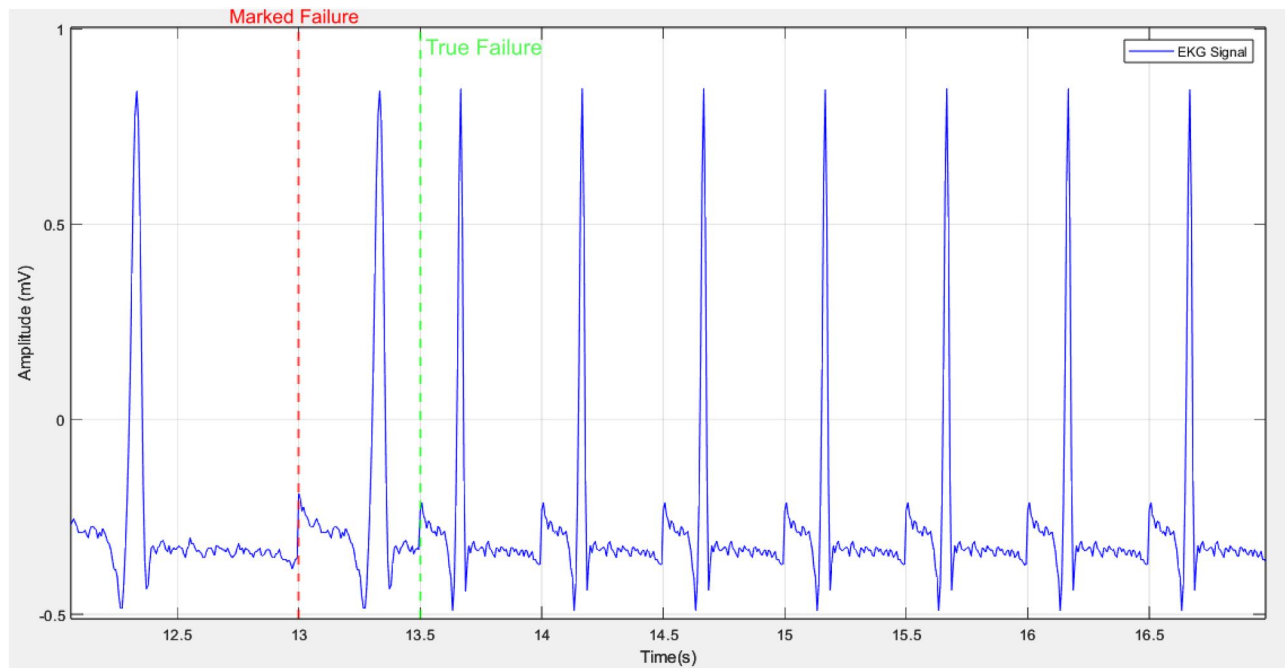


Figure 4. Timing discrepancy between the algorithm-detected marker and the actual anomaly onset.

corresponding to the last complete cycle before the actual onset. As illustrated in Figure 4, the actual anomaly begins at 13.5s (green line), and the detection marker is assigned to 13.0s (red line). While this strategy favors early flagging and robustness in real-time scenarios, it may reduce temporal precision in retrospective analyses or applications requiring exact anomaly onset localization.

To further validate the proposed method under practical conditions, an additional test was conducted using real EKG data from the PhysioNet MIT-BIH Arrhythmia Database (Goldberger *et al.* 2000; Kaggle 2025). This dataset contains patients' clinically recorded and annotated EKG signals, enabling assessment under authentic signal variability and noise conditions.

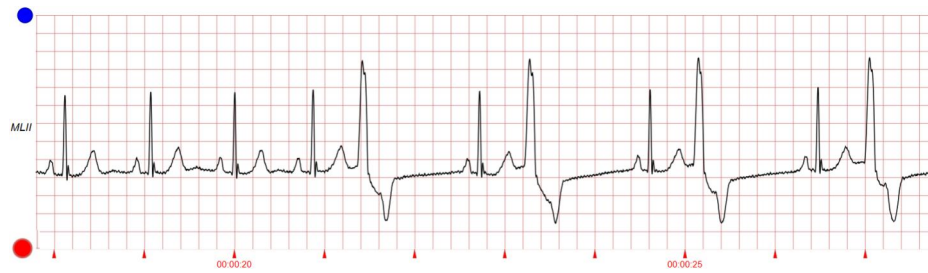


Figure 5. Original EKG signal from record 109 of the MIT-BIH Arrhythmia Database, displayed using the PhysioNet Lightwave viewer (Goldberger *et al.* 2000). The segment shown corresponds to approximately 18–28 s.

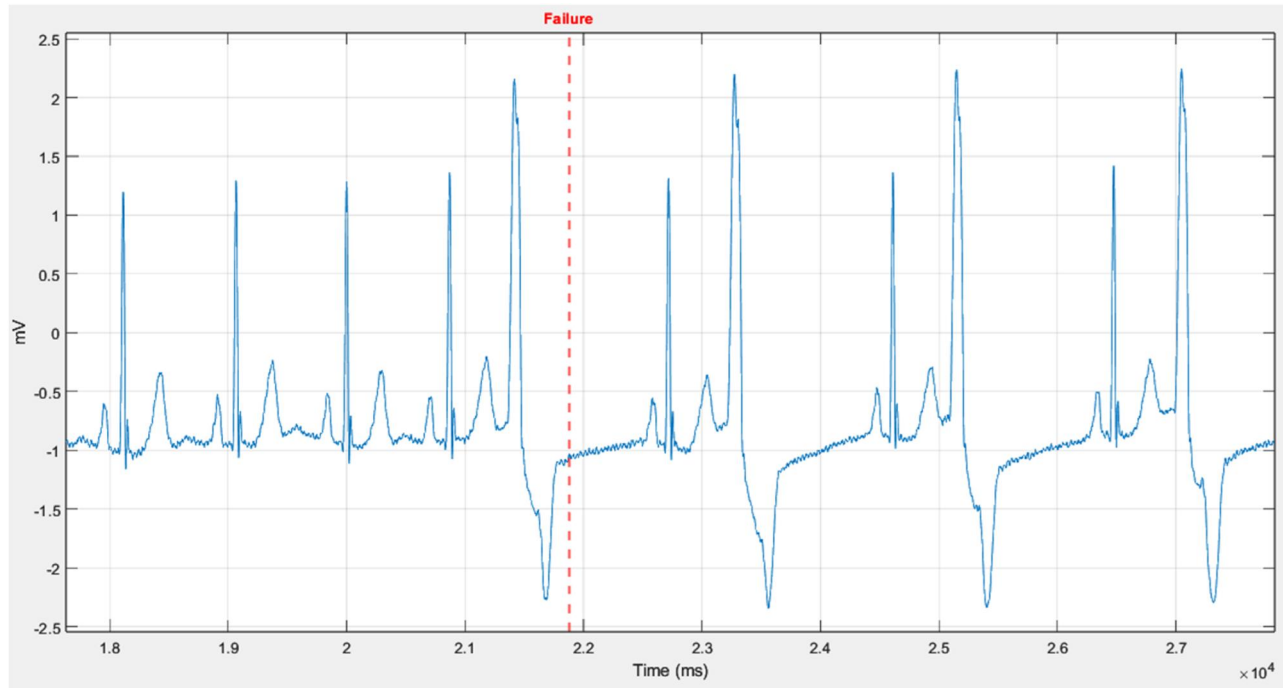


Figure 6. Output of the proposed detection algorithm for the same time interval.

Table 1. Confusion matrix with associated performance metrics.

	Actual positive	Actual negative	
Predicted positive	TP = 266	FP = 0	Sensitivity 95%
Predicted negative	FN = 14	TN = 20	Specificity 100%
	Precision 100%	F1 Score 97.44%	Accuracy 95.33%

The analyzed segment corresponds to a 51-year-old female patient diagnosed with atrial fibrillation. A 10-second segment (from 18 to 28 s) was selected for evaluation. Figure 5 shows the original EKG signal, illustrating a sustained arrhythmia episode. Although the clinical diagnosis for this patient is atrial fibrillation, this part of the signal shows a maintained ventricular rhythm disturbance, suggesting accelerated idioventricular rhythm.

The proposed sliding window algorithm was applied directly to the raw data. In this approach, the sliding windows move forward step by step

across the signal, comparing each portion until an anomalous cycle is detected. Figure 6 shows the output of the detection algorithm. Red dashed lines indicate the identified failure, which is marked approximately 0.45 s after the change in amplitude, as determined by the window alignment strategy. While clinical annotations could not identify the anomaly in this test, visual inspection confirmed that the detection aligned with the signal's rhythm irregularities. These results complement the synthetic tests previously described.

Table 1 presents the confusion matrix used to evaluate the performance of the proposed algorithm. A detection was considered a true positive if the algorithm generated at least one anomaly marker before or at the start of the second immediately preceding the actual onset of the anomaly. This criterion prioritizes early detection over precise timing. Missed

Table 2. Parametric comparison of DSW against similar methodologies. (best cases are highlighted)

Method	Accuracy	Sensitivity	Precision	F1 score	Recognition time	Threshold adjustment
DSW	95.33%	95%	100%	97.44%	0.03 sec	On the fly
Linear SVM	95.25%	94.25%	95.5%	94.25%	—	Fixed
Random Forest	96.25%	96.25%	96.25%	96.25%	—	Fixed
DLTWFR (Bagul et al. 2026)	98.2%	98.2%	98.1%	98.05%	1.23 sec	Through training

detections within this window were counted as false negatives. For standard signals, any detection was treated as a false positive. However, no such cases occurred in the experiment, indicating that all missed detections corresponded to control cases designed to evaluate the method's sensitivity limits.

These results present several performance metrics that compare the algorithm predictions against the ground truth to classify outcomes as true positives, false positives, true negatives, or false negatives. Each metric provides a different perspective on the system's effectiveness:

- Precision (100%) reflects the proportion of correct anomalies detected, ensuring that the algorithm minimizes false alarms.
- Sensitivity or Recall (95%) measures the algorithm's ability to detect real anomalies, indicating how consistently it responds to relevant events.
- Specificity (100%) quantifies how well the algorithm avoids false detections when no anomaly is present, reinforcing its robustness in clean signals.
- F1-score (97.44%) balances precision and recall into a single metric, providing an overall measure of detection reliability.
- Accuracy (95.33%) shows the overall proportion of correct classifications for all cases.

The algorithm demonstrated an outstanding performance in detecting both morphological and frequency-related anomalies, achieving a detection rate of 95% for waveform variations and 94.3% for frequency deviations. However, the selected configuration limits the algorithm's ability to capture more subtle or gradual transitions (5% for morphological differences and approximately three beats per minute for frequency changes). For instance, in a hypothetical scenario where heart rate increases progressively over long intervals until reaching a critical or visibly abnormal state, the algorithm would likely fail to flag these incremental changes as anomalies; this reflects a trade-off between avoiding false positives and maintaining sensitivity to small, progressive shifts in signal behavior.

A real-time simulation scenario was implemented using the MATLAB numeric computing environment

to evaluate the detection module's computational efficiency. The average execution time during the tests ranged from 0.02 to 0.04 s, with an average random-access memory (RAM) usage of 14 to 15.5 MB using a 12th generation Intel Core i3-1215U (1.20 Ghz) with 8 Gb of RAM. To establish a point of comparison with other established methods, Table 2 summarizes the measured results of DSW against well established methods, as linear support vector machine (linear SVM) and Random Forest (Bhangale *et al.* 2024), and well as recent proposals as deep learning-based two-way feature representation (DLTWFR) (Bagul *et al.* 2026).

From the comparison, it can be seen that DLTWFR outperforms DSW in accuracy, sensitivity and reliability (F1 score), but a much higher computational cost. DLTWFR takes about 40 times longer to inform about the occurrence of an event. Also, the algorithm takes 1889s to be trained, making very difficult to online adjust the threshold. This computational cost increases because DLTWFR divides the input information into blocks. The greater the number of partitions, the better the performance with the price of a slower system. On the other hand, DSW excels for their precision, flexibility and low computational cost.

Despite its strong performance and adaptability, DSW exhibits a major disadvantage: the algorithm first detects an anomaly, and takes a couple of clock cycles to inform this occurrence. For applications where delay is critical, a dual-set point approach would be recommended. The first set point aims to detect anomalies with significant morphological deviations beyond a critical point, triggering an alarm immediately without requiring multiple confirmations. Using finer sub-window recurrent analysis or real-time peak-tracking methodologies, a second set point may enhance temporal accuracy without compromising robustness.

5. Conclusions

The proposed three-comparison-confirmation criterion effectively eliminates false positives. It introduces an average delay of approximately three cycles (plus the computational latency) between the actual onset

of the anomaly and the moment the system flags it. While this delay affects the triggering of an alarm, the anomaly log remains unaffected, allowing for accurate further analysis. In practical terms, even for critical events, this short delay does not compromise response effectiveness, as human reactions typically exceed the delay introduced by the algorithm.

Additionally, due to the way window alignment is handled, the timestamp associated with the detection may appear slightly earlier than the actual onset of the event. For clinical analysis, this second-level misalignment is rarely critical, as the presence and nature of the anomaly are far more relevant than the exact onset time. Even for critical events, this minor delay does not compromise the overall responsiveness or utility of the system.

From a practical perspective, the algorithm executes in approximately 0.03 s and requires very little memory, demonstrating its suitability for real-time and resource-constrained platforms. Its minimal computational footprint makes it a viable candidate for wearable devices and even implantable systems such as pacemakers, where power and memory are minimal.

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Ethical approval

The data used in this study were obtained from the publicly available PhysioNet MIT-BIH Arrhythmia Database (Goldberger *et al.* 2000). For practical implementation, preprocessed CSV files derived from this database were downloaded from Kaggle (Kaggle 2025), facilitating direct numerical analysis in MATLAB without requiring third-party software for data conversion. All data were publicly available and used in compliance with relevant legal and ethical regulations. We followed the principles of data legitimacy, privacy protection, transparency of data sources, and ethical review in our use. By following these ethical norms, we ensured the morality and legality of the study and the credibility and social value of the findings.

Consent to participate

This study did not involve human participants recruited by the authors. All data used were publicly available and anonymized, obtained from open-access repositories for academic research. Therefore, individual consent was not required.

Consent for publication

The data used in this study were obtained from the publicly available MIT-BIH Arrhythmia Database (PhysioNet) and from Kaggle, where preprocessed versions of the same data were available for practical implementation. All data are publicly available and fully anonymized. Therefore, consent for publication was not required.

Author contributions

CRedit: **Samuel A. Torres-de-Anda**: Conceptualization, Data curation, Investigation, Methodology, Validation, Writing – original draft; **Luis F. Cisneros-Sinencio**: Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Writing – review & editing; **Alejandro Díaz-Sánchez**: Resources, Software, Supervision, Writing – review & editing; **Rubén Salas-Cabrera**: Conceptualization, Supervision, Visualization, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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References

- Aishah S. 2024. Advanced signal processing techniques for real-time systems in edge computing. *J Comput Signal Syst Res.* 1(1):20–29. <https://doi.org/10.70088/0c99tw16>
- Al-Shrouf A. 2020. ECG beat classification using a sliding window and correlation of the three-bit linear prediction

- error signal. *WSEAS Trans Signal Process.* 16(9):12–18. <https://doi.org/10.37394/232014.2020.16.2>
- Bagul SG et al. 2026. Deep learning-based two-way feature representation of the ecg images for cardiovascular disease detection. *Arab J Sci Eng.* 51(9):11921–11937. <https://doi.org/10.1007/s13369-025-10881-9>
- Bayvel P. 2013. Digital signal processing. In: Kaminow IP, Li T, Willner AE, editors. *Optical fiber telecommunications*. 6th ed. Academic Press; p. 123–150.
- Bera SC, Chakraborty B, Ray JK. 2005. A mathematical model for analysis of ECG waves in a normal subject. *Measurement.* 38(1):53–60. <https://doi.org/10.1016/j.measurement.2005.01.003>
- Bhangale K et al. 2024. Machine learning based heart disease prediction using ECG image. 2024 5th International Conference for Emerging Technology (INCET) <https://doi.org/10.1109/INCET61516.2024.10593572>
- Braverman V, Ostrovsky R. 2010. Effective computations on sliding windows. *SIAM J Comput.* 39(6):2113–2131. <https://doi.org/10.1137/090749281>
- Cedeño Pozo A, Trujillo Codorníu R. 2015. Ventanas deslizantes por bloques para la implementación online de la transformada wavelet discreta. *Comput Sist.* 18(4):767–772. <https://doi.org/10.13053/cys-18-4-1426>
- Chandola V, Banerjee A, Kumar V. 2009. Anomaly detection: a survey. *ACM Comput Surv.* 41(3):1–58. <https://doi.org/10.1145/1541880.1541882>
- Desale KS, Shinde SV. 2023. ECG signal processing using an optimized sliding window approach for concept drift detection and adaptation. *J Eng Sci Technol Rev.* 16(5): 180–185. <https://doi.org/10.25103/jestr.165.22>
- Goldberger AL et al. 2000. PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. *Circulation.* 101(23):e215–e220. <https://doi.org/10.1161/01.CIR.101.23.e215>
- Kachuee M, Fazeli S, Sarrafzadeh M. 2018. ECG heartbeat classification: a deep transferable representation. 2018 IEEE International Conference on Healthcare Informatics (ICHI)
- Kaggle. 2025. Getting started visualizing patient 100's ECG. Kaggle. <https://www.kaggle.com/code/protobioengineering/getting-started-visualizing-patient-100-s-ecg/input>
- Kehtarnavaz N. 2004. Real-time digital signal processing: based on the TMS320C6000. 1st ed. Newnes.
- Kulanuwat L et al. 2021. Anomaly detection using a sliding window technique and data imputation with machine learning for hydrological time series. *Water.* 13(13):1862. <https://doi.org/10.3390/w13131862>
- Malave AJ, Kaneshiro B. 2026. SENSI-EEG-Preproc-ICA-EKG-Trigger: A MATLAB framework for semi-automated identification of EKG and trigger artifacts in EEG using ICA and spectral characteristics. Stanford Digital Repository. <https://doi.org/10.25740/pm515wq0875>
- Mule SR et al. 2025. AI-based wearable systems for detecting cardiovascular anomalies. 2025 Global Conference on Information Technology and Communication Networks (GITCON) <https://doi.org/10.1109/GITCON65266.2025.11377352>
- Patel I et al. 2021. Extraction of features from ECG signal. *Int J Curr Res Rev.* 13(08):103–109. <https://doi.org/10.31782/IJCRR.2021.13806>
- Tamalu N et al. 2023. Fault detection in transmission lines: a denial constraint approach. *Proceedings of the 38th Brazilian Symposium on Databases (SBBDD).*
- Zhang L, Lin J, Karim R. 2016. Sliding window-based fault detection from high-dimensional data streams. *IEEE Trans Syst Man Cybern, Syst.* 47(2):289–303. <https://doi.org/10.1109/TSMC.2016.2585566>
- Zhang Y, Wang H, Li X, Chen L, Liu J. 2023. Real-time digital signal processing implementation for in-beam PET of radiotherapy imaging in HIMM. *Nucl Instrum Methods Phys Res A.* 1045:167545.